Inside the Overlap: Perception and Inference in Medical MLLMs

Categorizing the shared image–text information, and what current reasoning work targets

Caf · discussion notes with Claude

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Summary

- The overlap $I(I;T)$ in the Venn figure is *not* solved supervision. Shannon overlap says the information is in the image; whether a model can *use* it depends on perception and knowledge. The relevant quantity is the $\mathcal{V}$-usable overlap.
- **Three tiers inside the overlap:** direct perceptual (generic vision suffices), expert perceptual (needs visual-concept expertise), and inferential (findings → diagnosis; needs declarative knowledge and often context C). Shortcut-redundancy with C is a cross-cutting axis.
- **General vs. medical:** general-domain work separates perception from knowledge/reasoning (OK-VQA, Sherlock, Prism); for small general VLMs reasoning is often the bottleneck. In medicine the evidence points the other way: chain-of-thought underperforms direct answering, and supplying expert descriptions or ROIs fixes it, so *expert perception* is the bottleneck.
- **Current medical reasoning work** mostly supervises the inferential tier, which LLMs largely handle already. Report-mined and case-text chains copy perceptual claims from text and present them as observations, which risks enlarging the text-only region. Outcome-only RL cannot tell tiers apart.
- **Guideline-grounded reasoning** (Med-PRM, MedGuideX, FaithMed, Dr-LLaVA, MMedExpert-R1, MedAgent-Pro) makes the *inference* edge verifiable, but guidelines are conditional on findings: the strongest ones are text-only and assume F is given. Guidelines verify inference, not perception.
- **Perception signals are emerging** in two forms: indirect image-dependence rewards (MCR's clean-vs-corrupted gap, Med-OPD's degraded-image evidence advantage), which are image-ablation PVI in disguise; and explicit grounding (IoU-gated factual rewards for CXR, BI-RADS concepts, MedTRACE's grounding and evidence verification, tool-based perception in MedAgent-Pro). None yet uses lesion-level counterfactuals or types steps by provenance.
- **Direction:** an oracle ladder to measure each tier's share of error, a step-level PVI audit of reasoning datasets, and provenance-typed reasoning with type-specific verification.

1 Why the overlap is not "solved"

The companion notes split report content into image-only (coverage gap), text-only (attribution gap), and shared regions. It is tempting to treat the shared region as clean supervision. It is not, for two reasons.

Shannon overlap is model-independent; usable overlap is not. $I(I;T)$ measures whether information about T is *present* in I. What training can exploit is the information *usable* by the model family $\mathcal{V}$ [1]:

$$I_{\mathcal{V}}(I \to Y) \;=\; H_{\mathcal{V}}(Y \mid \varnothing) - H_{\mathcal{V}}(Y \mid I), \qquad I_{\mathcal{V}}(I \to Y) \le I(I;Y) \text{ (up to family expressivity).}$$

Unlike Shannon information, $\mathcal{V}$-information can be *created* by computation: a model with the right perceptual concepts or knowledge extracts what another model cannot. So the overlap a given Med-MLLM effectively sees is smaller than the Venn diagram suggests, and how much smaller depends on what extraction requires.

Extraction requirements differ sharply within the overlap. "Is this a CT?" and "Is this stage IIIA disease?" are both answerable from the image in principle (given context), but the second needs expert perception, domain knowledge, and context. These belong in different categories.

2 Categorizing the overlap

2.1 Three tiers plus one cross-cutting axis

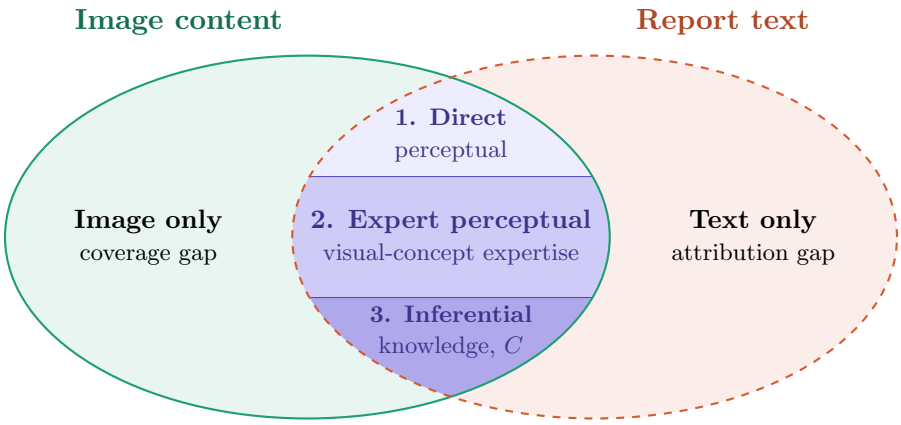


Figure 1: The overlap region subdivided by what extraction requires. Darker bands need more expertise or knowledge to turn present information into usable information. The bottom band borders the text-only region: many diagnoses are determined by findings *and* context C, so inferential content leaks into the attribution gap when C is absent.

Tier	Examples	What extraction needs	Status in Med-MLLMs
1. Direct perceptual	Modality, body part, view, gross mass, device presence	Generic vision; layperson-detectable	Mostly fine; inherited from general VLM pretraining
2. Expert perceptual	Ground-glass vs. consolidation, subtle pneumothorax line, mitotic figures, nodule margin	Visual-concept knowledge: *what the term looks like*; trained reader	Main bottleneck. A text LLM knows the word but not the appearance; reports supply the label without localization
3. Inferential	Findings → diagnosis, staging, differential, management	Declarative knowledge, often plus C (history, labs, priors)	LLM backbone largely has the knowledge (licensing-exam level); failures often trace to missing C or upstream tier-2 errors

Table 1: Tiers of the overlap region.

Cross-cutting axis: shortcut redundancy. Any tier can also be predictable from the question, the context, or the prior ("no pneumothorax" on a routine film). This is the redundancy case from the companion notes: the information is in the overlap, but the cheaper predictor is the prior. It is orthogonal to the tiers and must be controlled when measuring them (Section 5).

Operational definitions. To make the tiers measurable rather than intuitive:

- **Tier 1 vs. 2:** a finding is *direct* if non-experts (or a strong general-domain VLM zero-shot) detect it reliably; *expert* if detection requires a trained reader, measured by a layperson–radiologist agreement gap.
- **Tier 2 vs. 3:** the target is a *visual finding* (localizable in principle) vs. a *clinical conclusion* defined in terms of findings (and possibly C).

2.2 The perception–inference chain

Write the overlap as a chain $I \xrightarrow{\text{encode}} Z \xrightarrow{\text{perceive}} F \xrightarrow{\text{infer}} D$, with context C entering at the last step: $D = h(F, C)$. Tiers 1–2 live on the $Z \to F$ edge; tier 3 on the $F \to D$ edge. Errors propagate forward: an inference step cannot repair a tier-2 misread, and a chain-of-thought built on it can amplify it.

3 What the general domain already knows

- **Knowledge-requiring VQA.** OK-VQA [3] and A-OKVQA [4] separate questions answerable from the image alone from those needing outside knowledge, the analogue of tier 1 vs. tier 3.
- **Inference beyond literal content.** Sherlock [5] collects abductive inferences from visual clues: information entailed by the image but not directly visible.
- **Decoupling perception from reasoning.** Prism [6] has a VLM write a description and a separate LLM answer from that description alone. For small general VLMs, reasoning is often the bottleneck: pairing them with a stronger LLM raises scores, while perception stays roughly constant across LLM sizes.
- **Usable information.** $\mathcal{V}$-information [1] and pointwise $\mathcal{V}$-information [2] formalize "present but not usable".

What is missing: none of these frame direct vs. indirect as a property of the image–text overlap, and none has a tier corresponding to *expert perception*. In general images, perception is largely "direct"; the hard part is knowledge and reasoning.

3.1 Medicine reverses the bottleneck

- On medical VQA, chain-of-thought often scores *below* direct answering across general, medical, and closed models (e.g., Lingshu-7B 70.41 direct vs. 61.87 CoT; Hulu-Med-7B 76.42 vs. 62.09) [7]. The authors attribute this to a *medical perception bottleneck*: weak early grounding is amplified by the chain.
- Supplying ROI boxes and expert-level image descriptions raises CoT accuracy on SLAKE by 10–18 points [7]. Once the model is told what is in the image, reasoning works.
- MRPO's analysis of failure cascades in open-ended medical VQA finds that early invalid steps propagate and correlate with wrong answers; its manual taxonomy of 300 traces ties early-stage failures to *establishing the visual premise*, with mid-stage errors in interpreting perceived content [24].
- Our own token-zeroing result (concept-grounded cases near-empty; answers context-grounded) points the same way.

Hypothesis H_{overlap}

In medical MLLMs, the dominant share of error in the overlap region comes from tier 2 (expert perception), not tier 3 (inference). Equivalently: the perception gap Δ_{perc} in the oracle ladder below exceeds the inferential residual, especially for findings with a large layperson–expert gap.

4 What current medical reasoning work targets

Recipe	Examples	How chains are built	Tier trained	Gap effect
Report-mined chains	ChestX-Reasoner [8]	GPT-4o turns report findings/impressions into stepwise chains; unmentioned findings labeled normal	3, plus tier-2 claims copied from the report	Same $p(T \mid I)$ supervision, reformatted. Process reward checks observations against the *report*, not the image; the inference step is unchecked
Case-text distillation	Chiron-o1 [9]	Mentor LLMs reason over Radiopaedia case text (complaint, age, imaging captions, discussion)	3, with C inside the chain	Mentors told to present image observations as if seen directly; no step checked against the image. Context/report content relabeled as perception
Outcome-only RL	Med-R1 [10], MedVLM-R1 [11], GMAI-VL-R1 [12]	GRPO with format and answer-match rewards on multiple-choice VQA	Unclear: reward cannot separate tiers	Med-R1 No-Think matched or beat Think (72.57% vs. 71.14%): gains look like answer selection, not reasoning; also rewards question-text shortcuts
Perception-grounded reasoning	MedReason-R1 [13], CX-Mind [14]	Reasoning that zooms, crops, or interleaves image evidence	2	Targets the bottleneck; currently the minority

Table 2: Four early recipes for medical reasoning data, mapped to the tiers (see also the survey [15]). Section 4.1 positions more recent work.

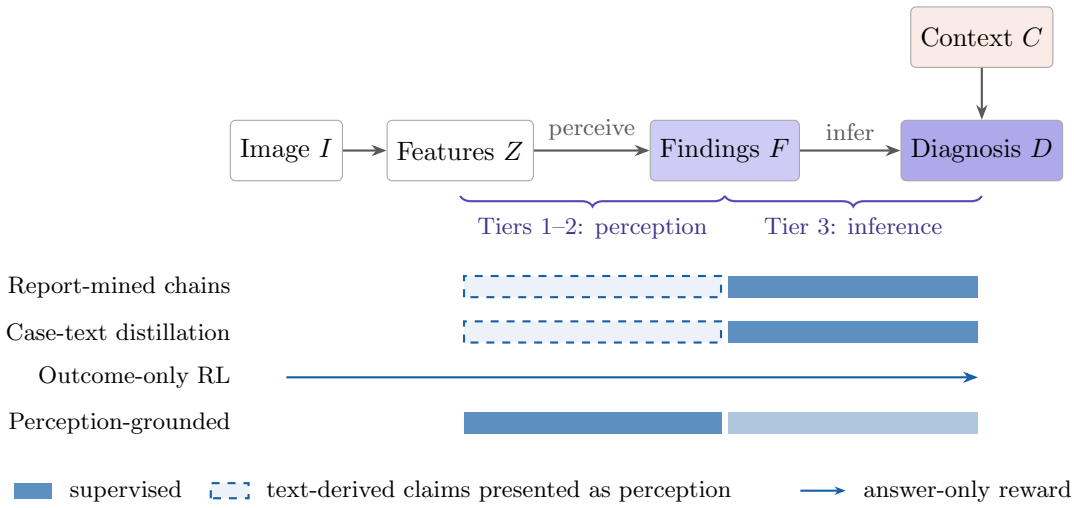


Figure 2: Which edge of the perception–inference chain each recipe supervises. Report-mined and case-text chains supervise inference well but cover perception only with text copied from reports or case descriptions (dashed), never checked against the image. Outcome-only RL rewards the end of the chain without saying which edge improved. Only perception-grounded methods put signal on the $Z \to F$ edge, where H_{overlap} locates the bottleneck.

4.1 Recent work: guidelines, process rewards, and perception signals

We position twelve recent papers along two axes: *which edge* of the chain they supervise, and *how* (if at all) claims about the image are verified. Tables 3–4 give details; Figure 3 gives the map.

Paper	Input	Structure source	Supervision / reward	Image verification
A. Guideline-verified inference, text only				edge $F \to D$
Med-PRM [17]	Text QA	RAG over guidelines, StatPearls, textbooks; LLM judge labels steps	Step-level PRM (±); best-of-N, rejection sampling	None (no image)
MedGuideX [19]	Text QA	CDC/PubMed guidelines → decision trees → executable functions	SFT on factual + counterfactual cases; GRPO with function-verified answers	None (no image)
FaithMed [21]	Text QA + retrieval	Clinician-designed evidence-based-medicine rubrics (Ask–Assess); citations	Outcome + step-level rubric process reward	None (no image)
B. Multimodal, structured reasoning; perception unverified				edge mostly $F \to D$
Dr-LLaVA [16]	Bone marrow pathology	Pathologist symbolic decision tree; templated dialogues	SFT + PPO: keyword correctness + rule-path consistency	None; labels only. Large consistency weight rewards valid-looking reasoning that ignores the image
MMedExpert-R1 [20]	Med VQA (mostly text samples)	GPT-5 traces under clinical-reasoning paradigm guidelines; specialty LoRA experts	SFT + RL (accuracy + format); TIES merging	None reported
MRPO [24]	Radiology, pathology VQA	MedThink gold rationales	GRPO: answer + step process reward vs. gold rationale; heavier penalty on early invalid steps	Steps judged against the rationale, not the image

Table 3: Recent medical reasoning papers positioned in the framework (part 1: groups A–B). "Image verification" asks whether claims about the image are checked against the image, as opposed to against reports, rationales, or labels.

Paper	Input	Structure source	Supervision / reward	Image verification
C. Indirect image-dependence signals				edge $Z \to F$ (implicitly)
MCR [18]	Med VQA (caption-derived)	Rejection-sampled CoT from a 72B model	GRPO: answer + chain-consistency judge + perceptual reward (clean vs. corrupted accuracy gap)	Indirect, batch-level image dependence
Med-OPD [25]	OmniMedVQA subsets	On-policy distillation, 4B → 2B	Token KL reweighted by evidence advantage (teacher log-lik. on original vs. degraded image)	Indirect, token-level; teacher sees the answer
D. Explicit perceptual grounding				edge $Z \to F$ (+ $F \to D$)
Spatial/factual RL [27]	CXR reports	Region-by-region traces; Chest ImaGenome boxes	SFT + GRPO: IoU spatial reward; RadGraph factual reward gated by IoU > 0.25	Explicit, boxes vs. annotations; reward hacking (duplicated findings) under spatial-only reward
MedCBR [23]	Breast US, mammography	BI-RADS concepts; guideline-prompted reasoning model	Concept, diagnosis, contrastive losses; reasoner frozen	Concepts supervised; diagnosis head reads the embedding, not the concepts
MedAgent-Pro [22]	10+ modalities	RAG (MedlinePlus) → guideline → indicator plan	Agentic: tools perceive (e.g., cup/disc segmentation → CDR), VLM infers	Explicit via expert tools, where tools exist
MedTRACE [26]	CXR+EHR, CXR, ECG	Hypothesis–tool–verify loop with evidence memory	Diagnosis, tool/evidence selection, 3-way verification, grounding (Dice/GIoU), calibration losses	Explicit grounding; evidence labeled supports / contradicts / insufficient

Table 4: Recent medical reasoning papers positioned in the framework (part 2: groups C–D).

	Text only: inference edge	Multimodal: whole chain	Multimodal: perception-targeted
Explicit grounding	n/a (no image)	MedTRACE, MedAgent-Pro (tools)	Spatial/factual RL (CXR), MedCBR (concepts) **★ gap: lesion counterfactuals + provenance typing**
Indirect image dependence	n/a (no image)	MCR (clean vs. corrupted)	Med-OPD (degraded-image advantage)
None / text-checked	Med-PRM, MedGuideX, FaithMed	Dr-LLaVA, MMedExpert-R1, MRPO, ChestX-Reasoner, Chiron-o1, Med-R1, MedVLM-R1	MedReason-R1, CX-Mind (zoom / interleave)

Figure 3: Positioning map. Columns: which edge of the chain is supervised. Rows: how claims about the image are verified. Shading darkens toward the cell that targets the tier-2 bottleneck with explicit verification. Guideline-grounded work concentrates in the bottom-left and bottom-middle: it verifies inference but not perception. The top-right cell is sparsely populated and is where the proposed provenance-typed reasoning sits. Placement of zoom/interleave methods reflects their design; their verification details were not examined here.

4.2 Interpretation

1. **Most data teaches the *shape* of clinical reasoning**, which the LLM backbone largely has. Perceptual steps come from report text, so the old $p(T \mid I)$ supervision returns in a new format, now framed as explicit "I see X" claims.
2. **Case-text chains are the riskiest.** History, labs, and discussion become "observed" claims: the text-only region converted into fake perception. This is a recipe for hallucinated findings and is consistent with CoT hurting medical VQA.
3. **Outcome-only RL improves use of the overlap but cannot block shortcuts**: it also rewards answers reachable from the question text alone, and No-Think ≈ Think suggests little of the gain is reasoning.
4. **Guidelines verify inference, not perception.** A guideline is a function $D = h(F, C)$: it presupposes the findings. MedGuideX makes h executable, so tier 3 becomes fully verifiable, and Med-PRM and FaithMed verify each inference step against retrieved evidence. But these strongest guideline-grounded methods are text-only. In multimodal settings (Dr-LLaVA, MMedExpert-R1) the guideline structure sits on top of unverified perception, and Dr-LLaVA notes that a large consistency weight rewards valid-looking reasoning that ignores the image: *inference supervision can crowd out perception.*
5. **"Faithfulness" is overloaded.** MCR's consistency reward checks that the answer follows from the chain; FaithMed's faithfulness checks evidence-based, cited reasoning. Neither checks faithfulness to the *image*. We suggest three distinct terms: *chain consistency* (tier 3), *evidence faithfulness* (text sources), and *image faithfulness* (tiers 1–2).
6. **Image-dependence signals are converging on PVI.** MCR's clean-vs.-corrupted accuracy gap and Med-OPD's teacher log-likelihood on original vs. degraded images are both image-ablation PVI estimates, at batch and token level. Two limits: generic degradation measures

dependence on *the image*, not on *the finding* (blur ≠ lesion removal), and Med-OPD's answer-aware teacher leaks the target into the signal. Lesion-level counterfactual edits are the clinically specific version.

7. **Explicit tier-2 verification is feasible but annotation-bound.** IoU-gated factual rewards, BI-RADS concept heads, and grounding losses all depend on region or concept annotations that exist for few modalities (Chest ImaGenome, BI-RADS, segmentation tools), and spatial-only rewards are hackable. MedCBR's diagnosis head bypasses its concepts, so its explanations need not be causal.
8. **Agentic decomposition externalizes perception.** MedAgent-Pro and MedTRACE assign perception to expert tools and inference to the VLM, matching the chain in Figure 2. This sidesteps the model's tier-2 weakness rather than fixing it, and only works where tools exist. MedTRACE's supports / contradicts / insufficient evidence labels are the closest precedent to provenance typing.

Caveat on expert perception. Experts often read images by gestalt rather than stepwise chains. Forcing verbose chains onto tier 2 may be the wrong shape; making perceptual evidence *explicit and checkable* (localization, counterfactual sensitivity) may matter more than making it long.

5 Measuring the tiers

5.1 A medical oracle ladder (Prism-style)

Evaluate the same diagnostic targets under four inputs, with loss (or error) L_k:

Rung	Input to the answering model	Gap isolated by the next rung
L_0	Image → Med-MLLM answer	$L_0 - L_1$: end-to-end vs. verbalized perception
L_1	Med-MLLM's own description $\hat{F}$ → LLM	$\Delta_{\text{perc}} = L_1 - L_2$: perception (tiers 1–2)
L_2	Oracle findings F → LLM	$\Delta_{\text{ctx}} = L_2 - L_3$: missing context
L_3	Oracle F + context C → LLM	L_3: inferential + intrinsic residual (tier 3)

Splitting Δ_{perc} by finding class (direct vs. expert, by the operational definitions above) separates tier 1 from tier 2. H_{overlap} predicts Δ_{perc} dominates and is concentrated in expert findings.

Controls. (i) Oracle F should come from structured, image-verified labels rather than free report text, which already contains context-influenced statements. (ii) Run a question-only baseline at each rung to subtract shortcut redundancy. (iii) Use counterfactual images (lesion removed/inserted) so that L_0 cannot be met by the prior.

5.2 Step-level PVI audit of reasoning datasets

For each step of a reasoning chain, compute image-ablation PVI (and context-ablation PVI where C is available) under a reference model. Prediction: "observation" steps in report-mined and case-text datasets have low image PVI, i.e., they are prior- or text-driven. If confirmed, this quantitatively shows that current medical reasoning data supervises the inferential and text-only regions, not the perceptual overlap.

6 Direction: provenance-typed reasoning

Tag each reasoning step by provenance and verify each type with the signal appropriate to it:

Step type	Tier	Verification / reward
Seen	1–2	Against the image: ROI/mask overlap; counterfactual sensitivity (remove the lesion and the step must change)
From context	bridges to text-only	Allowed only if C is in the input; otherwise abstain or request it
Inferred	3	Knowledge consistency given the *verified* findings; proper-scoring reward on stated confidence

- **Tier-2 supervision sources:** segmentation masks and ROI boxes (e.g., SLAKE), counterfactual lesion edits, atlas-style visual-concept data, and small amounts of expert localization.
- **Connection to latent awareness:** if Med-MLLMs internally encode whether an upcoming claim is image-driven (the "know–say" hypothesis from the companion discussion), provenance tags can be elicited from internal states and counterfactual PVI rather than taught from scratch.
- **Relation to the closest work.** Spatial/factual RL and MedTRACE show tier-2 verification is trainable; MCR and Med-OPD show image-dependence rewards help; MedGuideX shows tier 3 can be made executable. What none combines: (i) typing every step by provenance, including *from context* with abstention when C is absent; (ii) lesion-level counterfactuals instead of generic degradation; (iii) model-internal signals of image dependence. A natural composite: guideline-executable verification for *inferred* steps, grounding plus counterfactual sensitivity for *seen* steps.
- **Main risk:** the model learns to *label* steps without *looking* better. Evaluate on counterfactual pairs and on Δ_{perc}, not on tag accuracy alone.

7 Next steps

1. Define the tier split for a finding set (e.g., chest X-ray and CT findings) using layperson vs. radiologist detectability.
2. Run the oracle ladder on MedGemma and Lingshu with structured findings and counterfactual images; report Δ_{perc} (split by tier), Δ_{ctx}, and the inferential residual.
3. Step-level PVI audit of one report-mined and one case-text reasoning dataset.
4. Pilot provenance-typed reasoning with image-verified rewards for *seen* steps on a dataset with ROIs.
5. Re-evaluate representative positioned methods (e.g., MCR, MRPO, spatial/factual RL) under the oracle ladder and lesion-level counterfactuals, to test whether their gains come from the perception or the inference edge.

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